The electric motor heat exchange expression is

a. $C\frac{d\theta}{dt} + D\theta = p_1$

b. $p_1 + C\frac{d\theta}{dt} = -D\theta$

c. $p_1 + C\frac{d\theta}{dt} = D\theta$

d. $C\frac{d\theta}{dt} - D\theta = p_1$

Consumed power in induction motor is equal to

a. $P_1 = \sqrt{3} V_L I_{ph} cos\varphi_L$

b. $P_1 = \sqrt{3} V_{ph} I_{ph} cos\varphi_{ph}$

c. $P_1 = 3 V_{ph} I_L \ cos\varphi_{ph}$

d. $P_1 = 3 V_{ph} I_{ph} cos\varphi_{ph}$

The temperature limit of insulating materials class c is

Question 2

Answer

a.

150 °C

b.

> 180 °C

c.

130 °C

d.

180 °C

In the short time duty, the motor temperature

Question 3

Answer

a.

reaches the ambient temperature at rest period

b.

reaches the maximum temperature (equilibrium) at first period

c.

reaches the maximum temperature (equilibrium) at many periods

d.

doesn't reach the ambient temperature at rest period

In a half-wave rectifier, what happens to the negative half-cycle of the AC input?

Question 4

Answer

a.

It passes directly to the motor

b.

It is inverted to a positive cycle

c.

It is converted to positive DC voltage

d.

It is blocked and not used

A motor has a heating time constant of 60s and cooling time constant of 90s, if the motor rating is 10kw, find the permitted load power that can be used for 10s to let the motor work at its optimal utilization

Question 5

Answer

a.

25.5kw

b.

5kw

c.

10kw

d.

51kw

The ripple frequency in the DC output of a 3-phase full-wave rectifier fed by a 50 Hz AC source is

Question 6

Answer

a.
50 Hz

b.
300 Hz

c.
200 Hz

d.
150 Hz

The temperature rise is related to ambient temperature which value is equal to

Question 8

Answer

a.
20 °C

b.
40 °C

c.
30 °C

d.
10 °C

What happens to the DC motor speed when a freewheeling diode is added to the circuit?

Question 9

Answer

a.

Motor efficiency is reduced

b.

Speed increases due to reduced ripple

c.

Speed remains constant but torque improves

d.

Speed decreases due to increased ripple

Continuous duty cycle operates under constant load with following parameters: Motor duty is considered continuous duty (S1), if the motor operates with the following conditions

Question 10

Answer

a.
starting losses are not neglected

b.
temperature rise during operation under load does not reach steady-state value

c.
starting time is not neglected

d.
temperature rise during operation under load reaches steady-state value